Saujas Vaduguru

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Education

08/2022–present **Ph.D. in Language and Information Technologies**
Language Technologies Institute, School of Computer Science, Carnegie Mellon University
Advisor: Daniel Fried

08/2022–05/2025 **Master of Science in Language Technologies**
Language Technologies Institute, School of Computer Science, Carnegie Mellon University

07/2017–06/2022 **B.Tech. in Computer Science** and **M.S. (by Research) in Computational Linguistics**
International Institute of Information Technology (IIIT), Hyderabad
Advisors: Monojit Choudhury, Dipti Misra Sharma
Thesis: Program Synthesis for Linguistic Rules

Experience

05/2026–08/2026 **Research Intern**, *Microsoft Research*
Hosts: Chan Young Park, Jennifer Neville
Project: Collaborative multi-agent memory

05/2024–08/2024 **AI Research Scientist Intern**, *Autodesk*
Host: Yewen Pu
Project: Multi-turn interaction for design

2019–2022 **Research Assistant**, Language Technologies Research Center, *IIIT, Hyderabad*
Project: Program synthesis for linguistic rules

05/2021–08/2021 **Research Intern**, Chandar Research Lab, *Mila*
Mentors: Sarath Chandar, Prasanna Parthasarathi
Project: Continual learning for task-oriented dialogue

Publications

[13] ***Success and Cost Elicit Convention Formation for Efficient Communication***
Saujas Vaduguru, Yilun Hua, Yoav Artzi, Daniel Fried
64th Annual Meeting of the Association for Computational Linguistics (ACL), 2026

[12] **mrCAD: *Multimodal Refinement of Computer-aided Designs***
William P. McCarthy,* **Saujas Vaduguru**,* Karl D. D. Willis, Justin Matejka, Judith E. Fan, Daniel Fried, Yewen Pu
Findings of the Association for Computational Linguistics: EMNLP, 2025

[11] ***Identifying & Interactively Refining Ambiguous User Goals for Data Visualization Code Generation***
Mert Inan, Anthony Sicilia, Alex Xie, **Saujas Vaduguru**, Daniel Fried, Malihe Alikhani
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025

*equal contribution

[10] ***Analyzing Information Sharing and Coordination in Multi-Agent Planning***
Tianyue Ou, **Saujas Vaduguru**, Daniel Fried
Preprint, 2025

[9] **Is the Pope Catholic? Yes, the Pope is Catholic.** ***Generative Evaluation of Intent Resolution in LLMs***
Akhila Yerukola, **Saujas Vaduguru**, Daniel Fried, Maarten Sap
62nd Annual Meeting of the Association for Computational Linguistics (ACL), 2024

[8] ***Amortizing Pragmatic Program Synthesis with Rankings***
Yewen Pu, **Saujas Vaduguru**, Priyan Vaithilingam, Elena Glassman, Daniel Fried
41st International Conference on Machine Learning (ICML), 2024

[7] ***Generating Pragmatic Examples to Train Neural Program Synthesizers***
Saujas Vaduguru, Daniel Fried, Yewen Pu
12th International Conference on Learning Representations (ICLR), 2024

[6] ***Symbolic Planning and Code Generation for Grounded Dialogue***
Justin Chiu, Wenting Zhao, Derek Chen, **Saujas Vaduguru**, Alexander Rush, Daniel Fried
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023

[5] ***Adapting to Gradual Distribution Shifts with Continual Weight Averaging***
Jared Fernandez, **Saujas Vaduguru**, Sanket Vaibhav Mehta, Yonatan Bisk, Emma Strubell
Poster at the *Workshop on High-dimensional Learning Dynamics*
40th International Conference on Machine Learning (ICML), 2023

[4] ***Probing Negation in Language Models***
Shashwat Singh, Shashwat Goel, **Saujas Vaduguru**, Ponnurangam Kumaraguru
Poster at the *8th Workshop on Representation Learning for NLP (RepL4NLP), 2023*

[3] ***Stress Rules from Surface Forms: Experiments with Program Synthesis***
Saujas Vaduguru, Partho Sarthi, Monojit Choudhury, and Dipti Sharma
International Conference on Natural Language Processing (ICON), 2021

[2] ***Efficient Pragmatic Program Synthesis with Informative Specifications***
Saujas Vaduguru, Kevin Ellis, Yewen Pu
Contributed talk at the *Workshop on Meaning in Context: Pragmatic Communication in Humans and Machines, 2021*

[1] ***Sample-efficient linguistic generalizations through program synthesis: Experiments with phonology problems***
Saujas Vaduguru, Aalok Sathe, Monojit Choudhury, and Dipti Sharma
Proceedings of the 18th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology), 2021

Honors and Awards

2026 Microsoft Research Fellowship

2021 Mitacs Globalink Research Internship

2017–2021 Dean's List Award for Academic Performance, *IIIT, Hyderabad*

2015 Honourable Mention, *International Linguistics Olympiad*

Academic Service

Organizer: First Workshop on Theory of Mind at ICML 2023

Reviewer:* **NeurIPS 2025**, ICLR 2025, ACL Rolling Review (October 2025, May 2025, February 2025, June 2024, **August 2024**), EMNLP 2023, ACL 2023, BlackboxNLP@EMNLP 2023–2024, NILLI@EMNLP 2021

Teaching Experience

Fall 2026 **Teaching Assistant**, AI Agents, *Carnegie Mellon University*

Fall 2023 **Teaching Assistant**, Advanced Natural Language Processing, *Carnegie Mellon University*

Spring 2020 **Teaching Assistant**, Computational Linguistics I, *IIIT, Hyderabad*

Invited Talks

Spring 2026 Learning Efficient Communication in Multi-agent Interactions, *LeMN Lab, UC San Diego*

Fall 2024 Learning to synthesize programs from informative examples, *MPI Tübingen*

Fall 2023 Program synthesis from informative examples, *CMU Software Research Seminar*

Outreach

Pāṇini Linguistics Olympiad Team leader and coach for Indian team at the International Linguistics Olympiad (2018, 2019, and 2021), Co-chair of Problem Committee and Jury (2018–2021), Member of the Problem Committee (2022–)

Proficiencies

Programming languages Python, JavaScript

Frameworks Pytorch, HuggingFace Transformers

*Boldface indicates recognition as Outstanding Reviewer or Top Reviewer